Variation of Exhaust Noise with Engine Speed

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TOPICS WHICH WE WILL DISCUSS IN THE PRESENTATION

- INTRODUCTION
- THEORY
- OBJECTIVE
- PROCEDURE
- **SIMULATOR**
- **REFERENCES**

INTRODUCTION

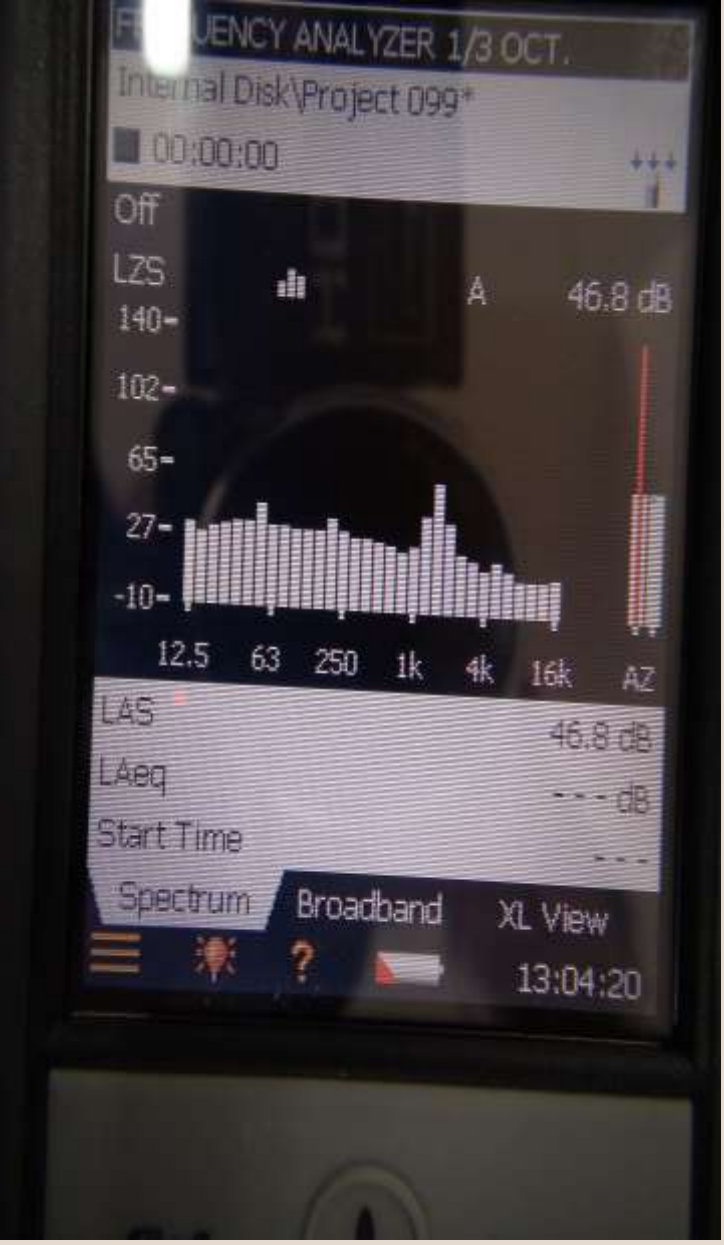
With every new machine, a little noise is created, with every mechanism employed to do man’s work, some mechanical or electrical power is converted into acoustical power, so that with the rise of people’s standard of living there occurs also a rise in the noise level of people’s confines. Noise control is becoming increasingly important for a wide variety of OEM designers. The internal combustion engine has mechanized the world. Since the early 1900s it has been our prime source of mechanical power. The vast number of internal combustion engines in the world today has resulted in air pollution, noise pollution etc.

The decibel (dB) is a logarithmic unit used to express the ratio between two values of a physical quantity, often power or intensity. One of these quantities is often a reference value, and in this case the decibel can be used to express the absolute level of the physical quantity, as in the case of sound pressure.

The instrument we are using for measuring sound levels is 2250 Light. Type 2250 Light is a modular Class 1 sound level meter measuring all standard acoustic parameters simultaneously, with frequency analysis and logging available in the optional application modules

THEORY

- **Various factors that contribute to the noise in engine**
- ***Combustion Noise:***
- Combustion noise is produced because of unsteady combustion of fluid and is of two types: turbulent combustion noise and periodic combustion oscillation.
- ***Mechanical Noise:***
- Mechanical noise is the noise which is generated by various impacts between the engine parts. This noise source is more important in the higher frequency range rather than in lower frequency range where combustion noise is important. There are lots of moving parts, for example, gear, valves, and rocker arms, piston and cylinder liner.
- ***Piston Slap Noise:***
- Piston slap noise is generated by the sudden impact of the piston to the cylinder wall is considered to be predominant due to the higher amount of energy released.
- ***Exhaust Noise:***
- The engine exhaust noise originates at the exhaust tailpipe openings and is transmitted through the cabin walls, firewall, and nose gear bay.This is quite high.


UENCY ANALYZER 1/3 OCT.
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46.8 dB
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FREQUENCY ANALYZER 1/3 OCT.
Internal Disk\Project 099*
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LZS
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34.7 dB
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Knocking

Knocking (also called knock, detonation, spark knock, pinging) in spark-ignition internal combustion engines occurs when combustion of the air/fuel mixture in the cylinder does not start off correctly in response to ignition by the spark plug, but one or more pockets of air/fuel mixture explode outside the envelope of the normal combustion front.

The fuel-air charge is meant to be ignited by the spark plug only, and at a precise point in the piston's stroke. Knock occurs when the peak of the combustion process no longer occurs at the optimum moment for the four-stroke cycle.

Types of Sound Levels

A-Weighting:

A-weighting filter covers the full audio range - 20 Hz to 20 kHz and the shape is similar to the response of the human ear at the lower levels - see the Equal Loudness Contours entry. A-weighted noise measurements are the most widely used and the Meter Class confirms the accuracy of the meter including the filters. The preferred convention is to write LA = x dB, however dBA and dB(A) are often used, as are LAeq, LAmax etc.

Z-Weighting:

Z for 'Zero' frequency weighting, which implies no frequency weighting. In reality the range is 10 Hz to 20 kHz ±1.5 dB.

Objective

To measure and monitor the noise radiated near the exhaust of the single cylinder SI engine as a function of engine speed.

PROCEDURE

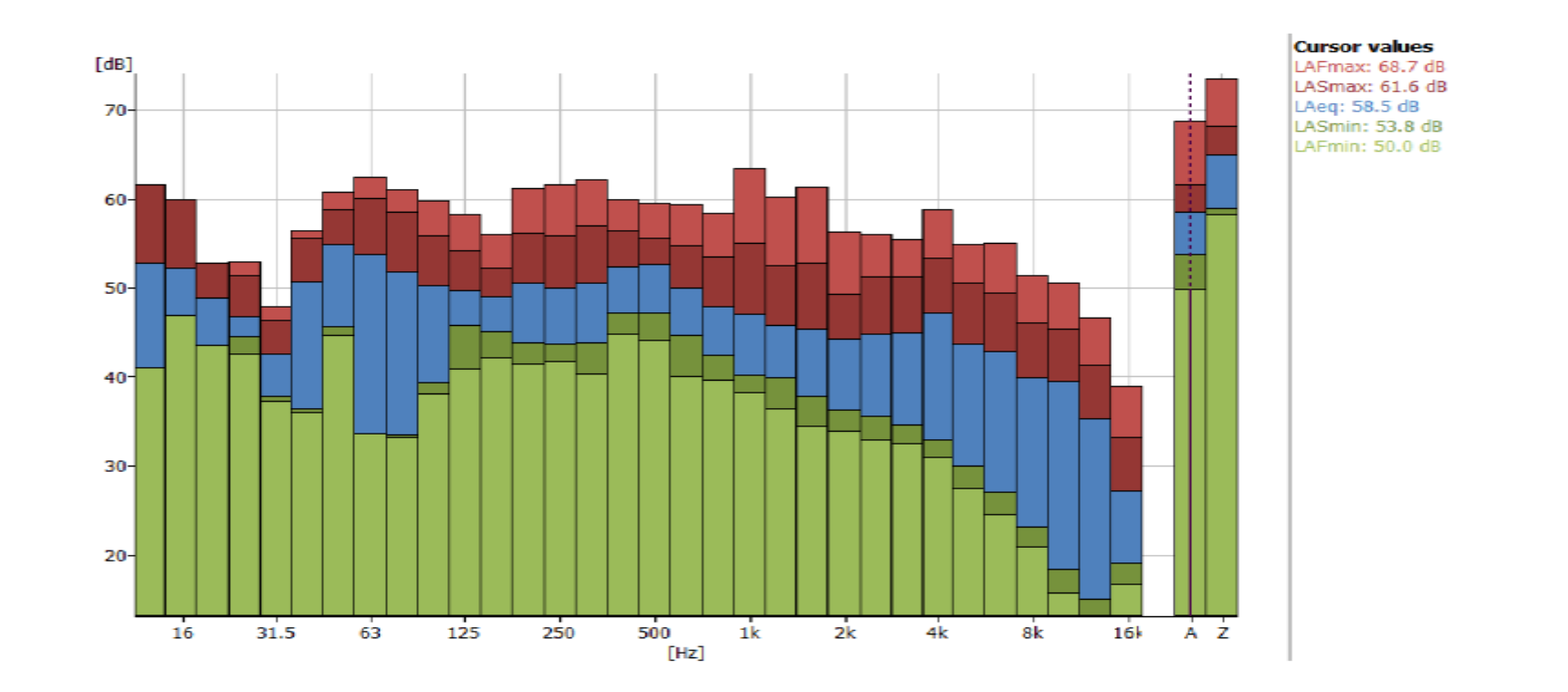
This remote triggered virtual lab is accessible over the internet you have to use the login user id and password provided to you when you registered for this experiment. Please refer to the detailed procedure sent to you by email. You need to have team viewer software installed in your computer with high speed internet access. You'll be able to get visual feed from the lab where the experimental setup is physically placed while the experiment is being conducted you have the option to access and retrieve the experimental data over the internet to your computer for further processing.

For working from the lab , one has to switch on the water pump, the motor, fuel supply, and the control panel.Once the computer starts, you need to open the already installed EPA software. Then you have to crank the engine. Adjust the dynamometer load control for varying the torque. After starting the engine,mount the Light 2250 (sound level meter) on a tripod stand and place it near the exhaust inside the room. Press the start button on the sound level meter . It will start recording data(signal).

- ***Precautions***
  1. Shoes are a must in the lab.
  2. The nut closing the opening of fuel pipe must be opened.
  3. No bubble must be present in the fuel tube.
  4. The water flow rate must be around 100.
  5. The emergency door must be kept open.

SIMULATOR

- **With engine not running**



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